

		mean power for square wave, $P_B = \frac{R}{2}$
29	D	For the spectral line with <u>shortest</u> wavelength, the energy difference between the involved levels = $\frac{hc}{\lambda_1}$, which is the <u>largest</u> For the spectral line with <u>longest</u> wavelength, the energy difference between the involved levels = $\frac{hc}{\lambda_2}$, which is the <u>smallest</u> Thus, the energy difference between levels for the third spectral line = $\frac{hc}{\lambda_1} - \frac{hc}{\lambda_2}$ Let the wavelength of this third spectral line be λ_3. Thus, the energy difference can also be expressed as $\frac{hc}{\lambda_3}$. We then have $\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2}$
30	B	Given $A = A_0 \exp(-\lambda t)$